

Solution: Winter Semester Exam 2022-23
GPD-508(Seismological Data Analysis)

Duration: 3 Hrs.

M.M. 100

Solution 1.

Output would be celtics and pacers.

It will display 76ers and celtics.

Solution 2.

a. *rotate to GCP*

Ans. It will rotate all components to Great circle path in radial transverse and vertical direction.

b. *sgf2ps f001.sgf sac_example.ps*

Ans. It will convert sac graphic format figure to human readable postscript file format.

c. *funcgen impulse delta 0.01 npts 100*

Ans. It will generate impulse functions having sample interval 0.01 and number of samples 100.

d. Merge

Ans. Used to concatenate two sac files.

e. Addf

Ans. Used to add two files.

Solution 3.

a. *Used to find arrival times of P-and S-waves arriving from 57.4 degrees and hypocentral depth of 200 km.*

b. *Taup_setsac -mod ak135 -evdpkm -ph P, S eqk_BHZ.sac*

c. *It will write lines between 30 and 60 to a file named "new"*

d.

Consider a file as follows (**A.txt**) that uses underscores to separate fields:

Vs_7.2
Vp_11.3
Rho_6.6

One could just extract the numeric values by:

```
>> cut -d_ -f2 A.txt
```

will

e. To extract columns of information from different files and combine them into one file

f. If two separate files share a common field they can be combined with join

Solution 4.

a.

The ray path γ is determined by the velocity structure using Snell's law & Ray Theory.

There are certain assumptions with Ray Theory

- <1> Ray in the vel. structure obey's Snell's Law
- <2> Rays should pass through shortest route in the velocity structure b/w source & receiver.
- <3> Travel-time depends only on velocity variations along the ray-path.

b. I will check the answer based on your understanding and learning about the bash shell scripting for guess the number between 0 to 200.

c. Consults class notes.

d.

Model Resolution Matrix (R) ^{problem.}

- * How accurately is the value of an inversion parameter recovered!
- * How small of an object can be imaged.

— Model Resolution Matrix R:

$$m = G^{-1} d_{obs} = G^{-1} G m_{true} = R m_{true}$$

← estimated model
Resolution matrix
← model which we really want to find

— R can be thought of as an spatial filter that is applied to the true model to produce the estimated values.

- * often just main diagonal analyzed to determine how spatial resolution changes with position in image.
- * off diagonal elements provide the "filter function" for every parameter.

- e. Consult class notes.

Solution 5.

- a. Seismic interferometry involves the cross correlation of responses at different receivers to obtain the Green's function between these receivers. For the simple situation of an impulsive plane wave propagating along the X-axis, the cross correlation of the responses at two receivers along the x-axis gives the Green's function of the direct wave between these receivers.
- b. Ambient noise tomography is a way to get a subsurface image by using an ambient noise source. When we have multiple stations in an area, we can cross-correlate them to one another, resulting in an impulse wave. This impulse wave is called the Green's function.
- c. Refer to class notes.
- d. Refer to class notes.

Solution 6.

Dispersion may be regarded as important feature of surface wave; where different frequency components travel with variable velocity.

Phase Velocity: velocity of individual frequency component.

$$c = \frac{\omega}{k}$$

ω - Angular freq.
 k - wave number

Group Velocity: velocity of the entire energy packet.

$$u = \frac{\partial \omega}{\partial k} = \frac{d(\omega k)}{dk}$$

$$\Rightarrow u = c + k \frac{dc}{dk}$$

In homogeneous medium, Rayleigh waves don't show dispersion bcz all waves travel with same velocities.

In homogeneous medium, $dc = 0$

$$u = c + 0 \Rightarrow \boxed{u = c}$$

But in heterogeneous media $\boxed{u > c}$ due to addition part. So, dispersion occurs in heterogeneous media.

Solution 7.

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Rayleigh waves in homogeneous medium

- * Rayleigh waves are combination of P & SV waves that can exist at the top of homogeneous half space $z=0$ [free surface]

Particle motion of Rayleigh wave is in $x-z$ plane, we consider only P & SV waves, bcz they can satisfy the free surface boundary condition & don't interact with SH waves. The P & SV potential are:

$$\phi = A \exp [i(\omega t - k_x x - k_z z)]$$

$$\psi = B \exp [i(\omega t - k_x x - k_z z)]$$

To describe energy trapped near the free surface, two conditions must apply. The soln must both ensure that the energy must not propagate away from surface & satisfy free surface boundary conditions:

① For this purpose, $e^{-k_z z} \rightarrow 0$ if $z \rightarrow \infty$

② Traction at free surface must vanish.

To solve potential wave eqn using above two condⁿ:

$$c_x = \left(2 - \frac{2}{\sqrt{3}}\right) \beta = 0.92 \beta$$

Apparent
vel. of Rayleigh
wave along free surface

Shear
ve vel.

Apparent vel. of
Rayleigh wave along
free surface is
1.1 times than

→ At $z=0$, the displacement components are

$$\left. \begin{aligned} u_x &= 0.42 A k_x \sin(\omega t - k_x x) \\ u_z &= 0.62 A k_x \cos(\omega t - k_x x) \end{aligned} \right\}$$

To visualize the above two Poisson's Solid
 eq^{ns} at $z=0$; consider there is a particle
 at $x=0$

At $t=0$, u_z is a maximum & $u_x=0$.
 As time increases the x & z displacements
 combine to give counterclockwise or
 "retrograde" elliptical motion.

For Poisson Solid, the maximum vertical
 displacement of the surface is about
 1.5 times the maximum horizontal
 displacement.

Wave ...

slightly lower than
Shear Wave velocity.

The coeff. of potential are related as

$$B = A \left(\alpha - \frac{c_x^2}{\beta^2} \right) / 2\gamma_\beta \quad \text{where} \quad \gamma_\beta = \left(\frac{c_x^2}{\beta^2} - 1 \right)^{1/2}$$

Relationship b/w potentials & displacement are as follows:

$$u_x = \frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial z} \quad \& \quad u_z = \frac{\partial \phi}{\partial z} + \frac{\partial \psi}{\partial x}$$

So, for Poisson Solid

$$u_x = A k_x \sin(\omega t - k_x x) \left[\exp(-0.45 k_x z) - 0.56 \exp(-0.39 k_x z) \right]$$

$$u_z = A k_x \cos(\omega t - k_x x) \left[-0.45 \exp(-0.45 k_x z) + 1.47 \exp(-0.39 k_x z) \right]$$

Both displacements decays with depth w.r.t $e^{-k_x z}$, so the depth to which a Rayleigh has significant displacement is proportional to its horizontal wavelength.

